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Jia

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(54) **DIGITAL ASSISTANT CREATION**

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G06N 20/00 (2019.01)

(52) **U.S. Cl.**
CPC ***G06F 8/30*** (2013.01); ***G06N 20/00*** (2019.01)

(58) **Field of Classification Search**
CPC G06N 3/006
See application file for complete search history.

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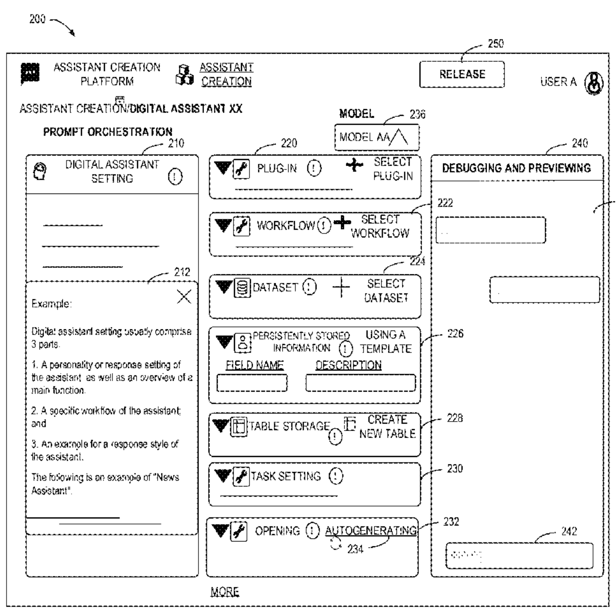
(57) **ABSTRACT**

Embodiments of the disclosure provides technologies for digital assistant creation. A method includes: in response to receiving a creation request, presenting a page for creating a digital assistant, the page comprising at least one configuration area for receiving configuration information for the digital assistant, the at least one configuration area comprising: a first configuration area for receiving settings information input in a natural language; and in response to receiving a release request, releasing the digital assistant based on the configuration information, for use in interaction with the user. Therefore, by providing a modular, simple-input scheme for digital assistant creation, users are able to define digital assistants easily and quickly with different capabilities without requiring code writing capabilities.

18 Claims, 8 Drawing Sheets



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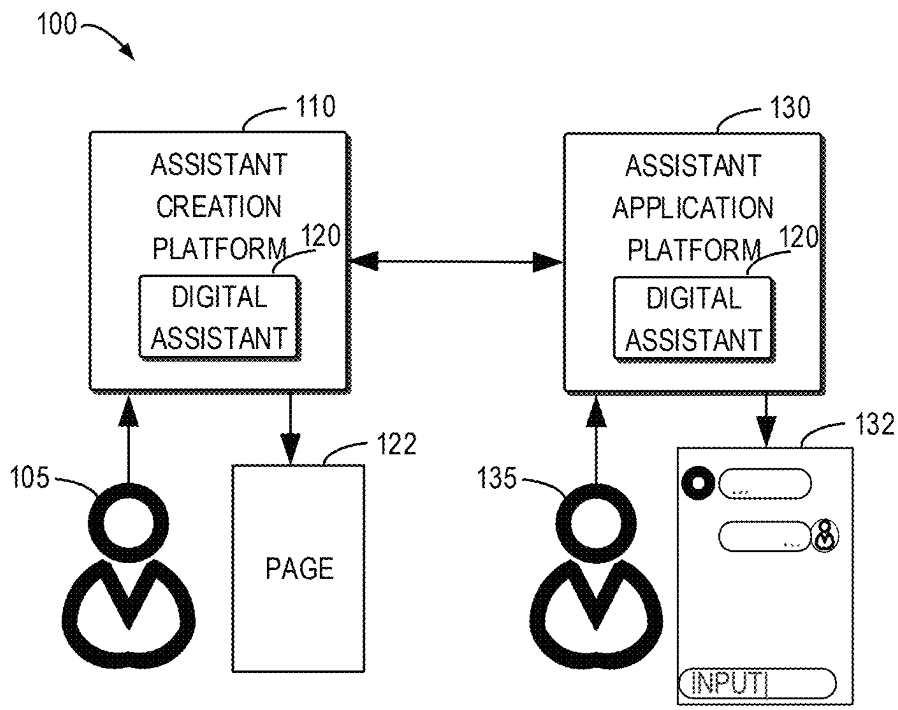
100
110
ASSISTANT
CREATION
PLATFORM
120
DIGITAL
ASSISTANT
130
ASSISTANT
APPLICATION
PLATFORM
120
DIGITAL
ASSISTANT
105
122
PAGE
135
132
INPUT

FIG. 1

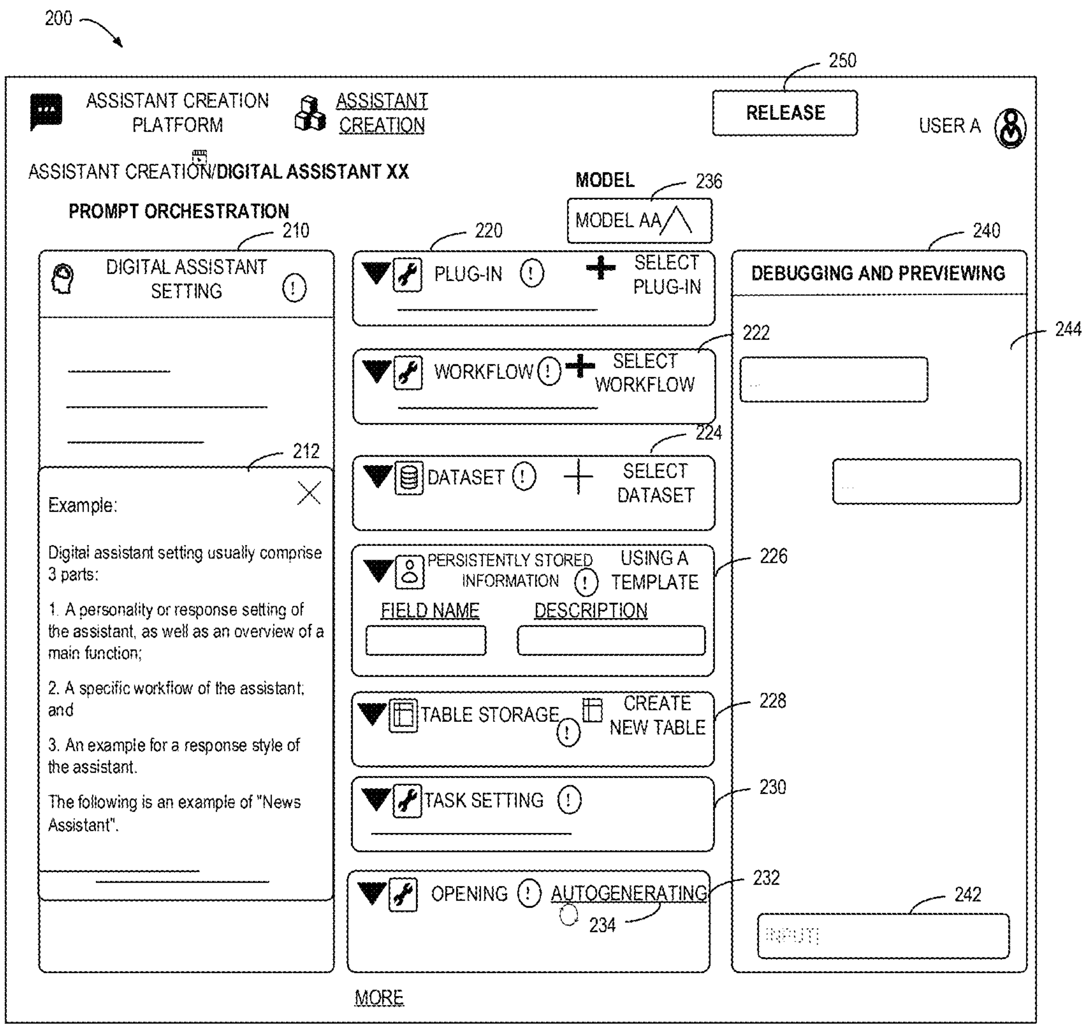
200
250
ASSISTANT CREATION PLATFORM
ASSISTANT CREATION
RELEASE
USER A
ASSISTANT CREATION/DIGITAL ASSISTANT XX
MODEL
236
MODEL AA
PROMPT ORCHESTRATION
210
220
240
DIGITAL ASSISTANT SETTING
PLUG-IN
SELECT PLUG-IN
DEBUGGING AND PREVIEWING
222
244
WORKFLOW
SELECT WORKFLOW
224
212
DATASET
SELECT DATASET
Example:
Digital assistant setting usually comprise 3 parts:
1. A personality or response setting of the assistant, as well as an overview of a main function;
2. A specific workflow of the assistant; and
3. An example for a response style of the assistant.
The following is an example of "News Assistant".
PERSISTENTLY STORED INFORMATION
USING A TEMPLATE
226
FIELD NAME
DESCRIPTION
TABLE STORAGE
CREATE NEW TABLE
228
TASK SETTING
230
OPENING
AUTOGENERATING
232
234
242
MORE

FIG. 2

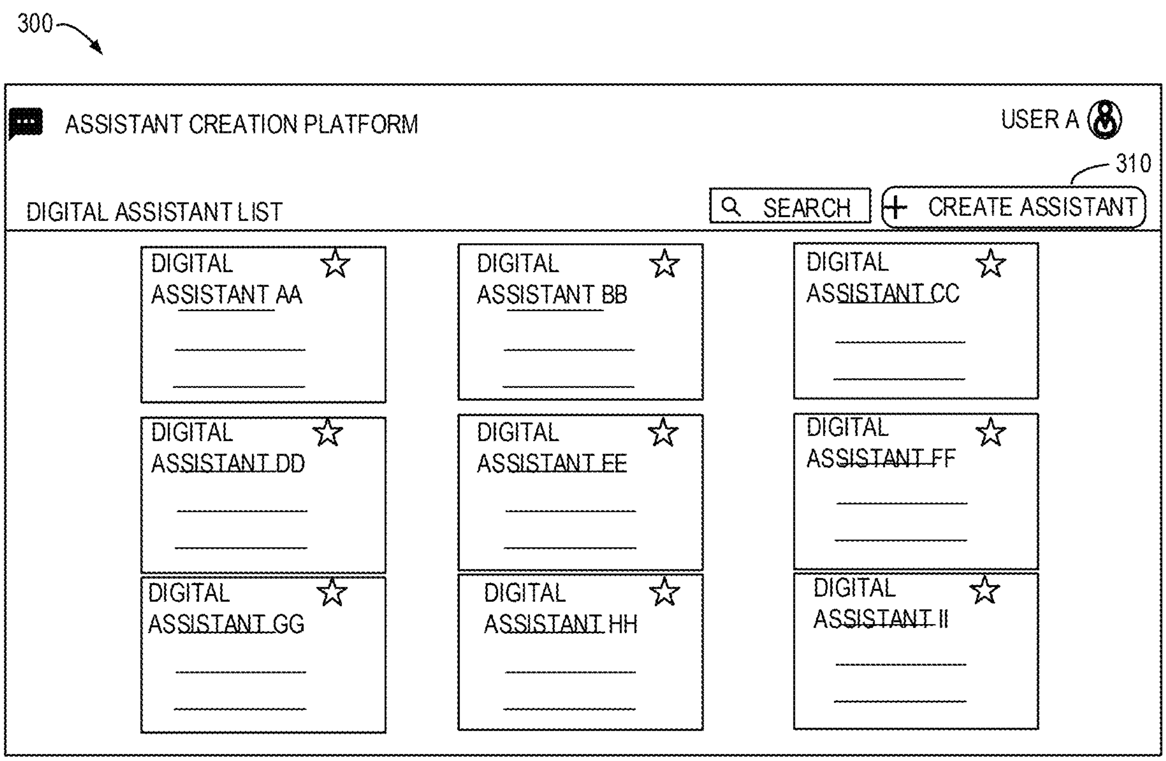
300
ASSISTANT CREATION PLATFORM
USER A
310
DIGITAL ASSISTANT LIST
SEARCH
CREATE ASSISTANT
DIGITAL ASSISTANT AA
DIGITAL ASSISTANT BB
DIGITAL ASSISTANT CC
DIGITAL ASSISTANT DD
DIGITAL ASSISTANT EE
DIGITAL ASSISTANT FF
DIGITAL ASSISTANT GG
DIGITAL ASSISTANT HH
DIGITAL ASSISTANT II

FIG. 3A

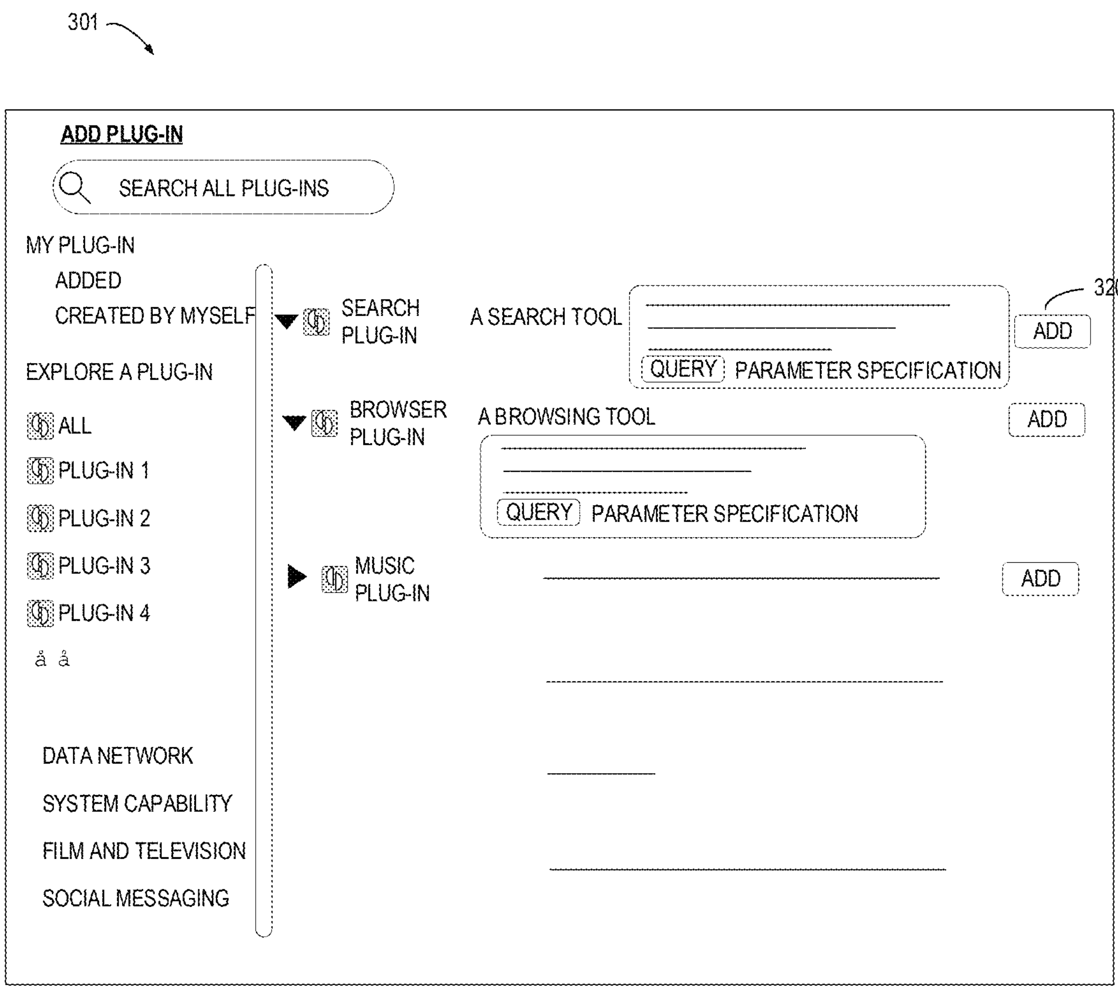
301
ADD PLUG-IN
SEARCH ALL PLUG-INS
MY PLUG-IN
ADDED
CREATED BY MYSELF
EXPLORE A PLUG-IN
ALL
PLUG-IN 1
PLUG-IN 2
PLUG-IN 3
PLUG-IN 4
DATA NETWORK
SYSTEM CAPABILITY
FILM AND TELEVISION
SOCIAL MESSAGING
SEARCH PLUG-IN
A SEARCH TOOL
QUERY
PARAMETER SPECIFICATION
320
ADD
BROWSER PLUG-IN
A BROWSING TOOL
ADD
QUERY
PARAMETER SPECIFICATION
MUSIC PLUG-IN
ADD

FIG. 3B

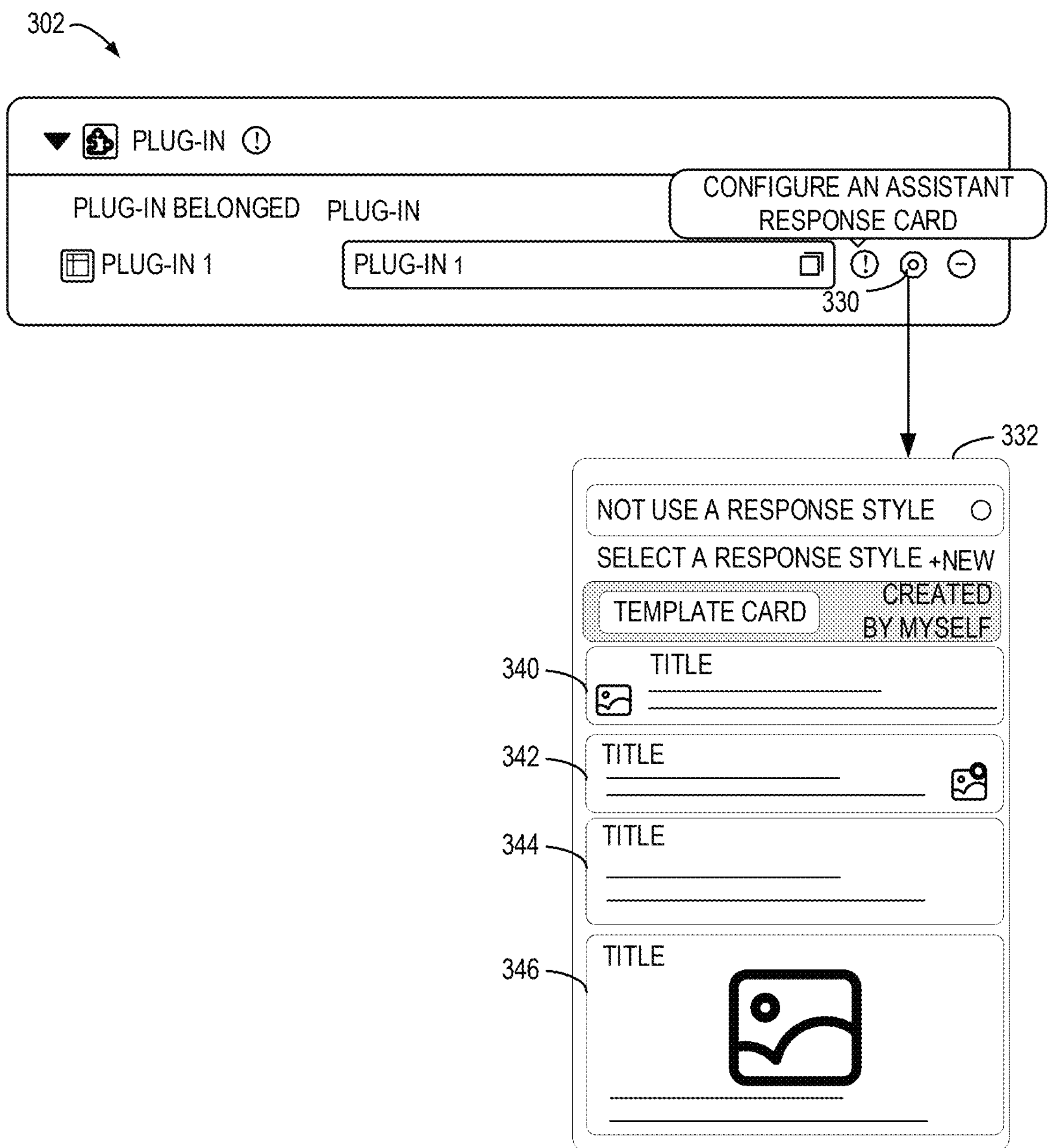
302
PLUG-IN
PLUG-IN BELONGED
PLUG-IN
CONFIGURE AN ASSISTANT RESPONSE CARD
PLUG-IN 1
PLUG-IN 1
330
332
NOT USE A RESPONSE STYLE
SELECT A RESPONSE STYLE +NEW
TEMPLATE CARD
CREATED BY MYSELF
340
TITLE
342
TITLE
344
TITLE
346
TITLE

FIG. 3C

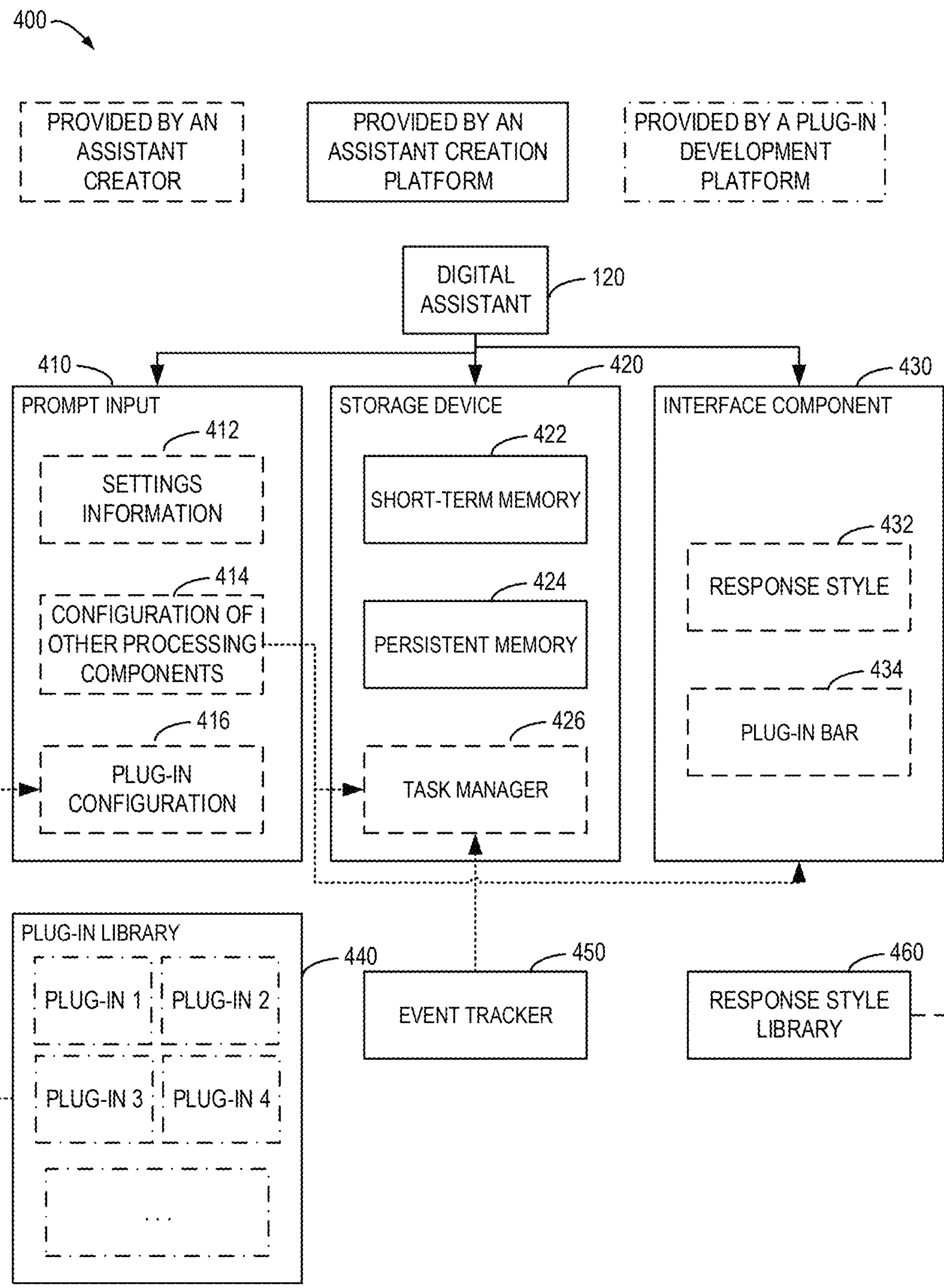
400
PROVIDED BY AN ASSISTANT CREATOR
PROVIDED BY AN ASSISTANT CREATION PLATFORM
PROVIDED BY A PLUG-IN DEVELOPMENT PLATFORM
DIGITAL ASSISTANT
120
410
PROMPT INPUT
412
SETTINGS INFORMATION
414
CONFIGURATION OF OTHER PROCESSING COMPONENTS
416
PLUG-IN CONFIGURATION
420
STORAGE DEVICE
422
SHORT-TERM MEMORY
424
PERSISTENT MEMORY
426
TASK MANAGER
430
INTERFACE COMPONENT
432
RESPONSE STYLE
434
PLUG-IN BAR
PLUG-IN LIBRARY
PLUG-IN 1
PLUG-IN 2
PLUG-IN 3
PLUG-IN 4
...
440
450
EVENT TRACKER
460
RESPONSE STYLE LIBRARY

FIG. 4

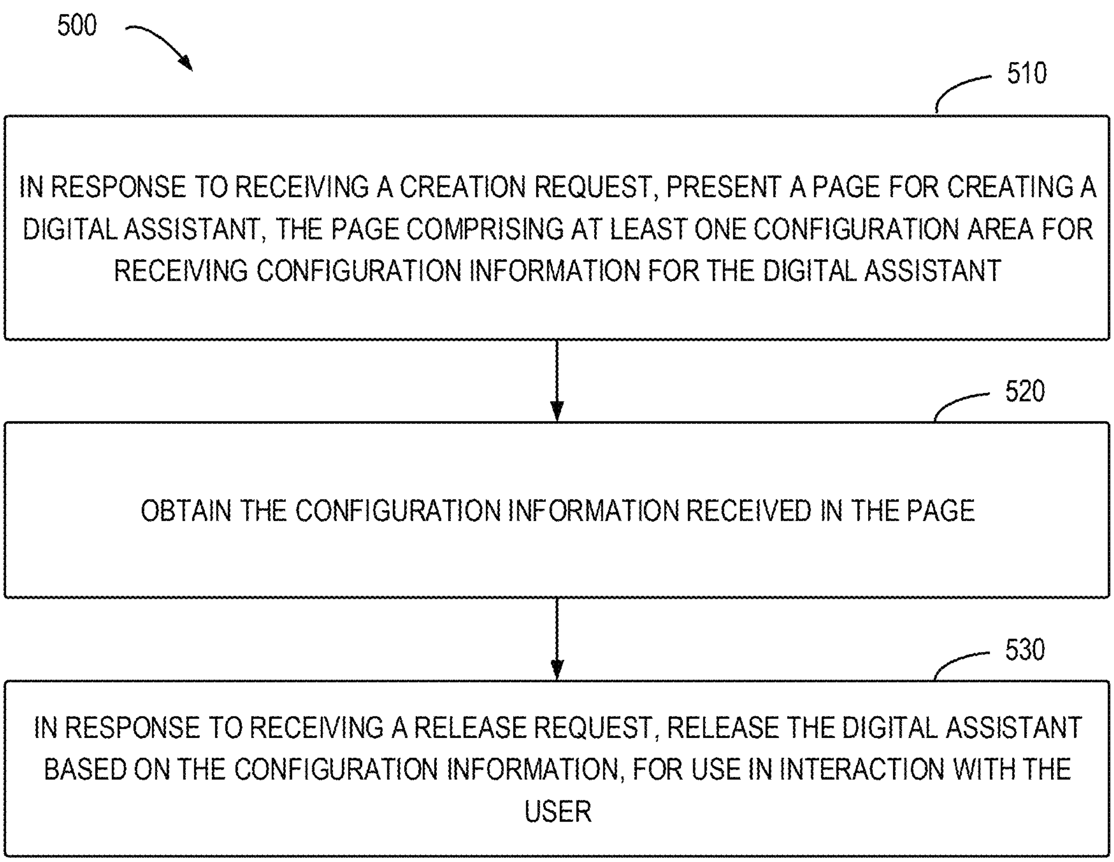
500
510
IN RESPONSE TO RECEIVING A CREATION REQUEST, PRESENT A PAGE FOR CREATING A DIGITAL ASSISTANT, THE PAGE COMPRISING AT LEAST ONE CONFIGURATION AREA FOR RECEIVING CONFIGURATION INFORMATION FOR THE DIGITAL ASSISTANT
520
OBTAIN THE CONFIGURATION INFORMATION RECEIVED IN THE PAGE
530
IN RESPONSE TO RECEIVING A RELEASE REQUEST, RELEASE THE DIGITAL ASSISTANT BASED ON THE CONFIGURATION INFORMATION, FOR USE IN INTERACTION WITH THE USER

FIG. 5

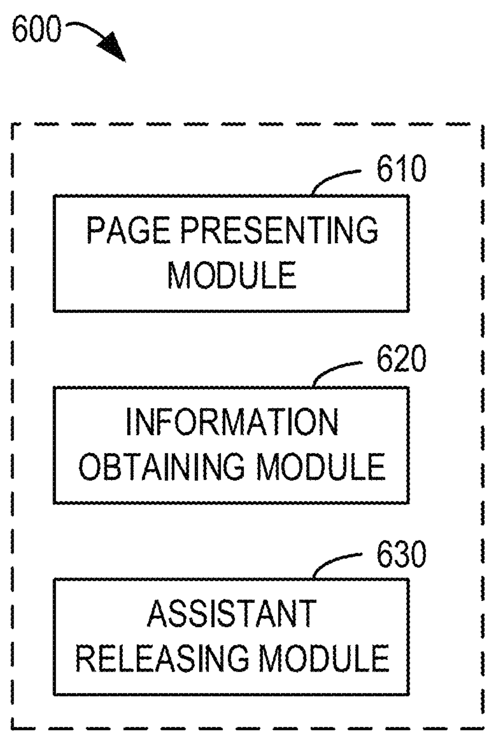
600
610
PAGE PRESENTING MODULE
620
INFORMATION OBTAINING MODULE
630
ASSISTANT RELEASING MODULE

FIG. 6

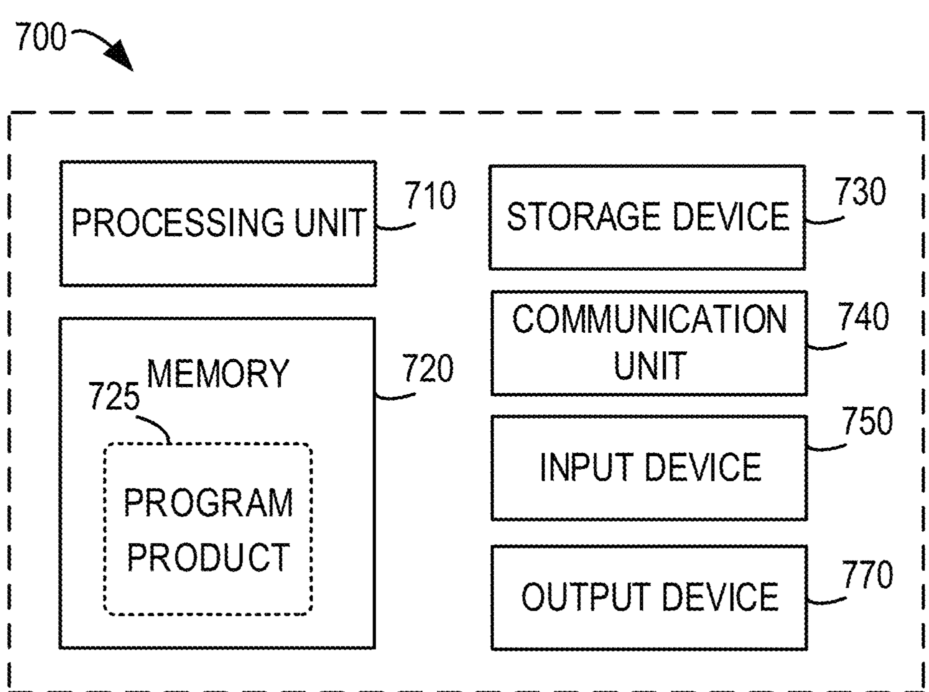
700
PROCESSING UNIT
710
MEMORY
725
PROGRAM PRODUCT
720
STORAGE DEVICE
730
COMMUNICATION UNIT
740
INPUT DEVICE
750
OUTPUT DEVICE
770

FIG. 7

DIGITAL ASSISTANT CREATION

CROSS-REFERENCE TO RELATED APPLICATIONS

The present application claims priority to Chinese Patent Application No. 202311415879.1, filed on Oct. 27, 2023, and entitled "METHOD, APPARATUS, DEVICE AND STORAGE MEDIUM FOR DIGITAL ASSISTANT CREATION", the entirety of which is incorporated here by reference.

TECHNICAL FIELD

The example embodiments of the present disclosure relate generally to the field of computers, and, more particularly, to digital assistant creation.

BACKGROUND

Digital assistants are provided to assist users in various task processing needs in different applications and scenarios. Digital assistants usually have intelligent dialogue and task processing capabilities. In the process of the interaction with digital assistants, users input interactive messages, and digital assistants respond to user input to provide response messages. Typically, digital assistants can support user inputs providing questions in a natural language format, and perform tasks and provide responses based on the understanding of the natural language input and logical reasoning capability of the digital assistants. Digital assistant interaction has become a useful tool that people love and rely on due to their flexible and convenient characteristics.

SUMMARY

In a first aspect of the present disclosure, a method for digital assistant creation is provided. The method comprises: in response to receiving a creation request, presenting a page for creating a digital assistant, the page comprising at least one configuration area for receiving configuration information for the digital assistant, the at least one configuration area comprising: a first configuration area for receiving settings information input in a natural language, the settings information being used to generate a prompt input of a machine learning model, and a response of the digital assistant to a user being determined by the digital assistant based on an output of the model; obtaining the configuration information received in the page; and in response to receiving a release request, releasing the digital assistant based on the configuration information, for use in interaction with the user.

In a second aspect of the present disclosure, an apparatus for digital assistant creation is provided. The apparatus comprises: a page presenting module configured to, in response to receiving a creation request, present a page for creating a digital assistant, the page comprising at least one configuration area for receiving configuration information for the digital assistant, the at least one configuration area comprising: a first configuration area for receiving settings information input in a natural language, the settings information being used to generate a prompt input of a machine learning model, and a response of the digital assistant to a user being determined by the digital assistant based on an output of the model; an information obtaining module configured to obtain the configuration information received in the page; and an assistant releasing module configured to, in response to receiving a release request, release the digital assistant based on the configuration information, for use in interaction with the user.

In a third aspect of the present disclosure, an electronic device is provided. The device comprises: at least one processing unit; and at least one memory, the at least one memory being coupled to the at least one processing unit and storing instructions for execution by the at least one processing unit. The instructions, when executed by at least one processing unit, cause the electronic device to perform the method of the first aspect.

In the fourth aspect of the present disclosure, a computer-readable storage medium is provided. The medium stores a computer program which, when executed by a processor, causes the device to perform operations that implement the method of the first aspect.

It would be appreciated that the content described in the section is neither intended to identify the key features or essential features of the present disclosure, nor is it intended to limit the scope of the present disclosure. Other features of the present disclosure will be readily understood through the following description.

BRIEF DESCRIPTION OF THE DRAWINGS

The above and other features, advantages, and aspects of the various embodiments of the present disclosure will become more apparent in combination with the accompanying drawings and with reference to the following detailed description. In the drawings, the same or similar reference symbols refer to the same or similar elements, wherein:

FIG. **1** shows a schematic diagram of an example environment.

FIG. **2** shows an example of a page for creating a digital assistant.

FIG. **3**A shows an example of a page for triggering the creation of a digital assistant.

FIG. **3**B shows a schematic diagram of an example interface for plug-in selection.

FIG. **3**C shows a schematic diagram of an example interface for configuring a response style.

FIG. **4** shows an example of a composition of a digital assistant constructed.

FIG. **5** shows a flowchart of an example process of application processing.

FIG. **6** shows a schematic block diagram of an example apparatus for application processing.

FIG. **7** shows a block diagram of an example electronic device.

DETAILED DESCRIPTION

The embodiments of the present disclosure will be described in more detail below with reference to the accompanying drawings. Although certain embodiments of the present disclosure are shown in the drawings, it would be appreciated that the present disclosure can be implemented in various forms and should not be interpreted as limited to the embodiments described herein. On the contrary, these embodiments are provided for a more thorough and complete understanding of the present disclosure. It would be appreciated that the accompanying drawings and embodiments of the present disclosure are only for the purpose of illustration and are not intended to limit the scope of protection of the present disclosure.

In the description of the embodiments of the present disclosure, the term "comprising", and similar terms would be appreciated as open inclusion, that is, "comprising but not limited to". The term "based on" would be appreciated as "at least partially based on". The term "one embodiment" or "the embodiment" would be appreciated as "at least one embodiment". The term "some embodiments" would be appreciated as "at least some embodiments". Other explicit and implicit definitions may also be included below.

Unless expressly stated herein, performing a step "in response to A" does not mean that the step is performed immediately after "A", but may comprise one or more intermediate steps.

It will be appreciated that the data involved in this technical solution (comprising but not limited to the data itself, data acquisition or use) shall comply with the requirements of corresponding laws, regulations, and relevant provisions.

It will be appreciated that before using the technical solution disclosed in each embodiment of the present disclosure, users should be informed of the type, the scope of use, the use scenario, etc. of the personal information involved in the present disclosure in an appropriate manner in accordance with relevant laws and regulations, wherein the relevant user may comprise any type of rights subject, such as individuals, enterprises, groups.

For example, in response to receiving an active request from a user, a prompt message is sent to the user to explicitly prompt the user that the operation requested operation by the user will need to obtain and use the user's personal information, so that users may select whether to provide personal information to the software or the hardware such as an electronic device, an application, a server or a storage medium that perform the operation of the technical solution of the present disclosure according to the prompt information.

As an optional but non-restrictive implementation, in response to receiving the user's active request, the method of sending prompt information to the user may be, for example, a pop-up window in which prompt information may be presented in text. In addition, pop-up windows may also contain selection controls for users to choose "agree" or "disagree" to provide personal information to electronic devices.

It will be appreciated that the above notification and acquisition of user authorization process are only schematic and do not limit the implementations of the present disclosure. Other methods that meet relevant laws and regulations may also be applied to the implementation of the present disclosure.

As used in this specification, the term "model" can learn a correlation between respective inputs and outputs from training data, so that a corresponding output can be generated for a given input after training is completed. The generation of the model can be based on machine learning techniques. Deep learning is a machine learning algorithm that processes inputs and provides corresponding outputs by using multiple layers of processing units. A neural networks model is an example of a deep learning-based model. As used herein, "model" may also be referred to as "machine learning model", "learning model", "machine learning network", or "learning network", and these terms are used interchangeably herein.

Digital assistants can serve as effective tools for people's work, study, and life. In general, the development of digital assistants is similar to the development of general applications, requiring developers having programming skills to define the various capabilities of digital assistants by writing complex code, and deploying digital assistants on appropriate operating platforms so that users can download, install, and use digital assistants.

With the diversification of application scenarios and the increasing availability of machine learning technology, digital assistants may be developed with different capabilities to support task processing in various segmented fields or meet the personalized needs of different users. However, limited by programming capabilities and limited understanding of the underlying implementation logic of digital assistants, users cannot freely and conveniently create different digital assistants. Therefore, this specification describes technologies configured to provide more convenient and flexible ways to create digital assistants, allowing more users to configure the wanted digital assistants.

According to embodiments of the present disclosure, an improved scheme for digital assistant creation is provided. According to this scheme, a page for creating a digital assistant is provided, which comprises one or more configuration areas for receiving configuration information for the digital assistant. In this page, a first configuration area is used for receiving settings information input in a natural language, the settings information is used to generate a prompt input of a machine learning model, and a response of the digital assistant to a user is determined by the digital assistant based on an output of the model. In this way, for users who require the creation of a digital assistant, settings information can be quickly input at least on this page to complete the creation process of the digital assistant. Afterwards, based on the configuration information received in the page, the digital assistant can be released for interaction with the user.

Therefore, by providing a modular, simple and free-input digital assistant creation scheme, users can easily and quickly define digital assistants with different capabilities without requiring user coding capabilities.

FIG. **1** shows a schematic diagram of an example environment **100** in which embodiments of the present disclosure can be implemented. The environment **100** involves an assistant creation platform **110** and an assistant application platform **130**.

As shown in FIG. **1**, the assistant creation platform **110** can provide a creation and release environment of a digital assistant for a user **105**. In some embodiments, the assistant creation platform **110** can be a low-code platform that provides a collection of tools of digital assistant creation. The assistant creation platform **110** can support visual development for digital assistants, thereby allowing developers to skip the manual coding process and accelerate the development cycle and cost of applications. The assistant creation platform **110** can support any suitable platform for users to develop digital assistants and other types of applications, for example, it can comprise platforms based on application platform as a service (aPaaS). Such a platform can support users to efficiently develop applications, achieve application creation, application function adjustment, and other operations.

The assistant creation platform **110** can be deployed locally on the terminal device of the user **105** and/or can be supported by a remote server. For example, the terminal device of the user **105** can run a client (e.g., an application) in communication with the assistant creation platform **110**, which can support the user's interaction with the assistant creation platform **110**. In the case where the assistant creation platform **110** is run locally on the user's terminal device, the user **105** can directly use the client to interact with the local assistant creation platform **110**. In the case where the assistant creation platform **110** is run on a server device, the server-side device can implement the provision of services to the client executing on the terminal device based on the communication connection between the assistant creation platform and the terminal device. The assistant creation platform **110** can present a corresponding page **122** to the user **105** based on the operation of the user **105** to output and/or receive information from the user **105**.

In some embodiments, the assistant creation platform **110** may be associated with a corresponding database that stores data or information required for the process of digital assistant creation supported by the assistant creation platform **110**. For example, the database may store code and descriptive information corresponding to various functional modules that make up the digital assistant. The assistant creation platform **110** may also perform operations such as invoking, adding, deleting, updating, etc. on the functional modules in the database. The database may also store operations that can be performed on different functional blocks. Exemplary, in a scenario where a digital assistant is to be created, the assistant creation platform **110** may invoke corresponding functional blocks from the database to construct the digital assistant.

In some embodiments of the present disclosure, the user **105** may create a digital assistant **120** on the assistant creation platform **110** as needed and release the digital assistant **120**. The digital assistant **120** may be released to any suitable assistant application platform **130**, as long as the assistant application platform **130** can support the execution of the digital assistant **120**. After releasing, the digital assistant **120** may be used for conversational interaction with a user **135**. The client of the assistant application platform **130** may present an interaction window **132** of the digital assistant **120** in the client interface, such as the conversation window. For example, the client may render a user interface in the terminal device for presenting the interaction window. The digital assistant **120**, as an intelligent assistant, has intelligent conversation and information processing capabilities. The user **135** may enter a conversation message in the conversation window, and the digital assistant **120** may determine a reply message based on the created configuration information and present it to the user in the interaction window **132**. In some embodiments, depending on the configuration of the digital assistant **120**, the interaction message with the digital assistant **120** may comprise messages in various message formats, such as text messages (e.g., natural language text), voice messages, image messages, video messages, and so on.

The assistant creation platform **110** and/or the assistant application platform **130** may run on an appropriate electronic device. The electronic device may be any type of computing-capable device, comprising a terminal device or a server-side device. The terminal device may be any type of mobile terminal device, fixed terminal device, or portable terminal device, comprising mobile phones, desktop computers, laptop computers, notebook computers, netbook computers, tablet computers, media computers, multimedia tablets, personal communication system (PCS) devices, personal navigation devices, personal digital assistants (PDAs), audio/mobile player, digital cameras/video cameras, positioning devices, television receivers, radio broadcast receivers, e-book devices, gaming devices, or any combination thereof, comprising accessories and peripherals of these devices, or any combination thereof. Server-side devices may comprise, for example, computing systems/servers, such as mainframes, edge computing nodes, computing devices in cloud environments, and so on. In some embodiments, the assistant creation platform **110** and/or the assistant application platform **130** may be implemented based on cloud service.

It will be appreciated that the structure and function of the environment **100** are described for the purposes of illustration only, without implying any limitation on the scope of the present disclosure. For example, although FIG. **1** shows a single user interacting with the assistant creation platform **110** and a single user interacting with the assistant application platform **130**, but multiple users can actually access the assistant creation platform **110** to create a digital assistant respectively, and each digital assistant can be used to interact with multiple users.

In the following, some example embodiments of the present disclosure will be described in detail with reference to the accompanying drawings. It should be understood that the pages shown in the drawings are merely examples and various page designs may actually exist. The various graphic elements in the page may have different arrangements and visual representations, one or more of which may be omitted or replaced, and one or more other elements may also exist.

The process of digital assistant creation described in this specification can be implemented on the assistant creation platform, with the terminal device installed on the assistant creation platform and/or the server corresponding to the assistant creation platform. In the following examples, for the sake of discussion, the assistant creation platform **110** is described from the perspective of the assistant creation platform, e.g. the assistant creation platform **110** shown in FIG. **1**. The pages presented by the assistant creation platform **110** can be presented via the terminal device of the user **105**, and user input can be received via the terminal device of the user **105**. The user **105** who creates the digital assistant is sometimes referred to as an assistant creator, assistant developer, etc.

The user **105** can initiate a creation request to the assistant creation platform **110** as needed. In response to receiving the creation request, the assistant creation platform **110** presents a page for creating a digital assistant. On this page, the user **105** can configure the digital assistant to be created (for example the digital assistant **120** shown in FIG. **1**). Different from creating the digital assistant by writing code, in embodiments of the present disclosure, the page for creating a digital assistant is designed to comprise at least one configuration area for receiving configuration information for the digital assistant. Each configuration area is defined to receive a type of configuration information required for constructing the digital assistant. FIG. **2** shows an example of a page **200** for creating a digital assistant **120** in accordance with some embodiments of the present disclosure.

Specifically, the page comprises at least a first configuration area for receiving settings information input in a natural language. For example, the page **200** in FIG. **2** provides a configuration area **210**, which comprises an input block for receiving settings information input by a user in a natural language.

The received settings information, as part of the configuration information of the digital assistant **120**, will be used for generating a prompt input for a model, and a response of the digital assistant **120** to the user is determined by the digital assistant based on an output of the model. That is to say, the digital assistant **120** to be created will understand the user input with the assistance of the model and provide a response to the user based on the output of the model. The model used by the digital assistant **120** can run locally on the assistant creation platform **110** or on a remote server. In some embodiments, the model can be a machine learning model, a deep learning model, a learning model, neural networks, etc. In some embodiments, the model can be based on a language model (LM). The language model can have a question-answering capability by learning from a large corpus of data. The model can also be based on other appropriate models.

During the creation process, a specific configuration area is provided for users to provide settings information, and the configuration of settings information can be completed by the user based on a natural language input. This way, users can easily constrain the output of the model and configure diverse digital assistants.

In some embodiments, the page may further comprise a second configuration area for receiving a configuration of at least one processing component, the configuration of the processing component indicating at least one processing component that the digital assistant **120** is capable of using when processing a user request. In some embodiments, the configuration of the processing components, for the created digital assistant **120** and when interacting with a user, may be provided to the model. The model may determine which and/or what processing components need to be used to complete the processing of the user input, and thus determine a response for the user.

In some embodiments, in the page for creating a digital assistant, one or more processing components to be used can be pre-configured or recommended for the digital assistant to be created. For example, a creation entry for creating different types of digital assistants can be provided. For certain types of digital assistants, the processing components that digital assistants of that type usually need to use can be pre-configured or recommended. In this way, users do not even need to select processing components, but only need to input different settings information to obtain a customized digital assistant. The response style and format, certain workflows and functions of these digital assistants can be determined based on the settings information input by the user.

In the digital assistant, each processing component can be understood as a tool that the digital assistant **120** can invoke when processing user requests, and each processing component is able to perform corresponding functions or services. The types of processing components can be very diverse, and can be selected, configured, or modified by the user **105** from existing processing components, or can allow the user **105** to customize one or more processing components. As shown in FIG. **2**, the configuration area in page **200** for receiving the configuration for at least one processing component comprises configuration areas **220**, **222**, **224**, **226**, **228**, **230**, **232**, etc. The configuration of processing components will be described in more detail below.

By understanding user requests with the assistance of models and settings information, and performing the user requests with the assistance of processing components, the digital assistant **120** will be able to interact with users and respond to user requests. The page used to create a digital assistant can be templated to provide various configuration areas for receiving the configuration information of the digital assistant **120**. The user **105** can complete a customization of the digital assistant **120** without performing complex configurations and coding.

In some embodiments, the assistant creation platform **110** may provide a creation entry to the digital assistant in any suitable page. A user may access the page for creating a digital assistant by triggering the creation entry. FIG. **3**A shows an example of a page **300** for triggering a creation of a digital assistant **120** in accordance with some embodiments of the present disclosure. The page **300** may be, for example, a home page of the assistant creation platform **110**, which comprises a creation entry **310**. Based on a trigger operation of the creation entry **310**, the page **200** of FIG. **2** may be presented. FIG. **3**A represents one specific example, and in fact, other creation entries may be set in the assistant creation platform **110**.

Based on the input for creating a digital assistant of the user **105** in the page, configuration information received in the page can be obtained. The configuration information comprises at least the settings information received in the first configuration area. After completing the configuration, the user **105** is also allowed to release the created digital assistant. In response to receiving a release request, the assistant creation platform **110** releases the digital assistant **120** based on the configuration information received in the page for interaction with the user. As shown in FIG. **2**, page **200** presents a release control **250**. In response to detecting a trigger operation on the release control **250**, the assistant creation platform **110** receives the release request of the user and releases the digital assistant **120** based on the configuration information received on page **200**.

In some embodiments, the created digital assistant **120** may be released to run on a default platform. In some embodiments, a candidate platform may be provided for user selection. In response to receiving the release request, the assistant creation platform **110** may provide at least one candidate platform, each of the at least one candidate platform supporting the execution of the digital assistant **120**. In response to receiving a determination of a target platform amongst the at least one candidate platform, the digital assistant **120** is released to the target platform, e.g. the assistant application platform **130** in FIG. **1**.

In some embodiments, the settings information may indicate a definition of a response style of the digital assistant **120** to be created. By setting the response style, the responses of the created digital assistant can be differentiated and can exhibit a specific personality to the user. Alternatively, or in addition, in some embodiments, the settings information may indicate a description of a function supported by the digital assistant **120** to be created. For example, in the configuration area **210** of page **200**, the user **105** may be allowed to input a text string, e.g., "You are a movie commentator, please use sharp and humorous language to explain the movie plot and introduce newly released movies to the user." Such settings information can guide the response style of the digital assistant **120** (e.g., "sharp humor") and describe the functions of the digital assistant **120** (e.g., "movie commentary", "explaining the movie plot", and/or "introducing newly released movies").

In some embodiments, alternatively or additionally, the settings information may indicate at least one workflow to be performed by the digital assistant **120** to be created. Each workflow may correspond to individual operations of the digital assistant **120** while performing a particular function. That is to say, the user **105** may be allowed to describe, in a natural language format, how the digital assistant **120** is to perform a certain function.

In some embodiments, alternatively or additionally, the settings information may indicate at least one response format of the digital assistant **120** to be created. The response format may comprise, for example, Markdown (a lightweight markup language) and the like.

It should be understood that the above only provides some examples of the settings information, and embodiments of the present disclosure are not limited in this regard. In fact, because the settings information will be used to construct prompt input for the model, users are allowed to freely try different settings information to construct a digital assistant that meets their expectations. For example, in the settings information, users can be allowed to input requirements for the response language of the digital assistant **120** and constraint conditions on the response content of the digital assistant **120** (e.g., the number of words for different types of responses, the type of response content, etc.).

In some embodiments, to better guide the user to complete the configuration of the digital assistant **120**, a settings information example can also be provided in the page for guiding the user to provide settings information of the digital assistant **120**. The settings information example can be provided at a location associated with the first configuration area for receiving the settings information. As shown in FIG. **2**, a settings information example **212** can be provided near the configuration area **210**, which can indicate a general communication of the settings information of the digital assistant **120** to the user and can provide specific settings information of a certain digital assistant as an example.

In some embodiments, one or more types of processing components may be provided to select or configure options, which allows the user **105** to select, enable, or specifically configure as needed.

In some embodiments, the second configuration area may comprise an area for plug-in configuration, such as an area **220** shown in FIG. **2**. At least one plug-in used by the digital assistant **120** can be selected or customized by the user in this area. Each plug-in is configured to perform a respective function. For example, a search plug-in can perform a data search function; a browser plug-in can provide a webpage browsing function; a music plug-in can provide a music search and a playing function, and so on. Each plug-in can be considered as an atomic capability of the digital assistant **110**. The digital assistant **120** can invoke one or more plug-ins to process a user request. In some embodiments, the assistant creation platform **110** can provide a plug-in library from which the user **105** can select developed plug-ins. FIG. **3**B shows an example interface **301** for plug-in selection according to some embodiments of the present disclosure, in which the user can browse the plug-in list and select the plug-in they want to use. As shown in FIG. **3**B, an add control **320** for each plug-in can be provided to select the corresponding plug-in to be added to the digital assistant **120** by the user **105**. In some embodiments, alternatively or additionally, the assistant creation platform **110** may provide a plug-in definition interface to define a plug-in having a specific function as required by the user **105**.

In some embodiments, the second configuration area may comprise an area for workflow configuration, such as an area **222** shown in FIG. **2**. In this area, the user can select or customize at least one workflow to be performed by the digital assistant **120**. The workflow can not only be input in a natural language in the settings information, but the page **200** can also provide a workflow entry for the user **105** to select an existing workflow or define a workflow through a dedicated workflow design interface.

In some embodiments, the second configuration area may comprise an area for dataset configuration, such as an area **224** shown in FIG. **2**. At least one dataset may be selected by the user in the dataset configuration area, and the digital assistant **120** uses at least one dataset to determine a response to the user. The "dataset" may also be referred to as a "knowledge base". In determining a response to the user, the digital assistant **120** may retrieve the respective knowledge from the configured dataset for the response determination. In some embodiments, the assistant creation platform **110** may allow the user **105** to configure the dataset of the digital assistant **120** by selecting from an existing dataset, uploading a local dataset, or specifying an online dataset, etc.

In some embodiments, the second configuration area may comprise areas for configuring persistently stored information, such as areas **226** and **228** shown in FIG. **2**. Typically, during the interaction between the user and the digital assistant, due to reasons such as model input length, the digital assistant will extract limited contextual information from historical interactions to understand user input. However, for users interacting with the digital assistant, they may expect the digital assistant to maintain long-term memory of certain information, so as to continuously provide more targeted responses. If the digital assistant needs to repeatedly inquire about this information during the interaction with the digital assistant, it will cause a decrease in the user's experience. Therefore, in some embodiments of the present disclosure, the developer of the digital assistant is allowed to predefine at least one type of information that the digital assistant **120** needs to persistently store. With this configuration, the defined at least one type of information will be automatically extracted and stored for subsequent interactions during the interaction between the digital assistant **120** and the user. If the user updates a certain type of information during the interaction, the previously stored information will be overwritten or updated. The defined information will be persistently stored for the specific user interacting with the digital assistant **120**. In this way, from the perspective of the interactive user of the digital assistant **120**, the digital assistant **120** is able to remember or recall certain key information, thus providing a good interactive experience for the user.

In some embodiments, in area **226** of page **200**, the user can configure one or more types of information to be stored in an attribute-value pair. For example, the creator of the digital assistant can add field names and descriptions of the information to be stored in area **226**. In area **228** of page **200**, the user can configure a table to persistently store one or more types of information. In the table, the creator can define more and more complex structured data. By defining the information to be persistently stored, after the created digital assistant **120** is released for interaction with user **135**, the corresponding information entered by user **135** will be stored for a long time, e.g., beyond a current user interaction session with the digital assistant, and provided as interaction context to digital assistant **120** for subsequent interaction.

In some embodiments, the second configuration area may comprise an area for configuring tasks, such as area **230** shown in FIG. **2**. The configuration of tasks can be supported by a task plug-in. In area **230**, the user **105** may be allowed to configure the task plug-in to perform predetermined tasks or user-defined tasks. Through the configuration of the task plug-in, the digital assistant **120** can automatically execute predetermined tasks without specific triggering by interactive users or allow interactive users to create custom tasks as needed. In some embodiments, if the creator of the digital assistant has not made any configuration inputs in area **230** for configuring tasks, the task plug-in will not be included in the digital assistant **120**.

In some embodiments, since different plug-ins can perform different functions, the execution results can be fed back to the user interacting with the digital assistant **120**. To enrich the response style of the digital assistant **120**, response configuration controls can also be provided for at least one plug-in selected by the creator. The creator of the digital assistant **120**, i.e., the user **105**, can trigger the response configuration control to configure the response style of each plug-in corresponding to the function.

FIG. **3**C shows an example interface **302** for configuring response styles according to some embodiments of the present disclosure. A response configuration control **330** is provided at a position associated with each plug-in. Specifically, in response to detecting a trigger operation of the response configuration control for the first plug-in in at least one plug-in, multiple candidate response styles associated with the first plug-in are presented. In the example shown, each candidate response style can indicate a graphical user interface (GUI) style for presentation in association with the message provided by the digital assistant **120**. As shown in FIG. **3**C, the trigger operation of the response configuration control **330** will select a response style selection window **332**, which comprises multiple candidate response styles **340**, **342**, **344**, **346**. The user **105** can select the desired response style to use from these candidate response styles.

The user **105** who creates digital assistant **120** can select the target response style of the first plug-in from multiple candidate response styles. In this way, during the use of digital assistant **120**, if the first plug-in is invoked to perform the corresponding function, the result of the execution will be provided to the interactive user in the target response style.

In some embodiments, the page for creating a digital assistant may further comprise a third configuration area for receiving guidance information. In the example page **200** of FIG. **2**, an area **232** for receiving the guidance information is provided. The guidance information presented directly to the user **135** when the user **135** triggers interaction with the digital assistant **120**. In some embodiments, the guidance information comprises at least one of: description information for the digital assistant **120** and at least one recommendation of a question for the digital assistant **120**. The description information for the digital assistant **120** can provide the user **135** with a general understanding of the characteristics of the digital assistant **120**, the functions that can be implemented, etc. The recommendation question for the digital assistant **120** can inspire the user **135** to initiate interaction with the digital assistant **120**, or directly select the recommendation question to interact with the digital assistant **120**.

In some embodiments, the guidance information may be automatically generated. The information generation control may be provided in the page, for example, at a position associated with the third configuration area. In the example of FIG. **2**, an information generation control **234** may be provided near the area **232**. Since the description information of the digital assistant **120** and/or at least one recommendation of a question for the digital assistant **120** may be derived from the settings information, in some embodiments, in the process of creating the digital assistant **120**, in response to detecting a trigger operation of an information generation control associated with the third configuration area, candidate guidance information is generated based at least on the settings information received in the first configuration area. The candidate guidance information is presented in the third configuration area. In this way, the user **105** may determine whether to use the candidate guidance information, modify or replace the candidate guidance information, etc. as needed to obtain a final desired guidance information.

As described above, the digital assistant **120** can use a model to understand the user request and determine a response to the user. In some embodiments, the model used by the digital assistant **120** may be default and requires no configuration by the creator. In some embodiments, during the creation of the digital assistant **120**, the creator may be allowed to select the model to be used. A fourth configuration area may be provided in the page used to create the digital assistant for receiving a selection of the model. The selected model is invoked to determine a response of the digital assistant **120** to a user. As shown in FIG. **2**, the page **200** further comprises an area **236** for configuring the model, in which the user **105** may be allowed to select the model to be used.

The above discussed the configurable processing components in the process of digital assistant creation. In specific applications, the assistant creation platform can provide more, fewer, or different configurations of processing components as needed for the creator of the digital assistant to choose or configure.

In some embodiments, in order to enable the user **105** creating the digital assistant to easily test the execution effect of the created digital assistant **12** during the creation process, a debugging area for the digital assistant, such as a debugging area **240** shown in FIG. **2**, may also be provided in the page. The debugging area **240** comprises an input area **242** for receiving a debugging request for the digital assistant **120**, and a presentation area **244** for providing a debugging result for the debugging request (and providing the received debugging request). The debugging area **240** may be configured in the form of an interactive window, simulating an interactive interface viewed by an interactive user of the digital assistant **120**.

During the debugging process, the debugging results presented in the debugging area **240** can be determined based on the received debugging request and current configuration information for the digital assistant **120** in the page **200**. The user **105** can determine whether an actual execution result of the digital assistant **120** meets expectations based on the debugging result, determine whether to continue modifying the configuration information, or release the digital assistant. In some embodiments, for each debugging request, in addition to providing the debugging result, the digital assistant **120** may be provided to determine an underlying execution process of that debugging result, e.g., invocations of the model, the thought process of the model, one or more plug-in used, etc. This can allow the user **105** to more quickly determine whether the currently configured digital assistant meets expectations.

The above describes the process of creating a digital assistant in some embodiments of the present disclosure. In the embodiments of the present disclosure, the assistant creation platform provides sufficient support for the digital assistant constitution, so that users can easily, quickly, flexibly, and freely create the wanted digital assistant.

FIG. **4** shows an example of a composition of the digital assistant **120** constructed in accordance with some embodiments of the present disclosure. After the creator completes the configuration via the assistant creation platform **110**, the execution of the released digital assistant **120** is supported by the following three aspects: a prompt input **410**, a storage device **420**, and an interface component **430**.

Settings information **412** input by the creator, a plug-in configuration **416** input by the creator, and/or a configuration of other processing components **414** can be used together to determine the prompt input **410**, which will be provided to the model used by the digital assistant **120**. The plug-in configuration **416** can be selected from a plug-in library **440** provided by a plug-in development platform by the creator or can be added to the plug-in library **440** after being developed by the creator.

The storage device **420** of the digital assistant **120** may comprise a short-term memory **422** for storing short-term contextual information during interaction with the user, and a persistent memory **424** for storing one or more types of information to be persistently stored, defined in the configuration information input by the creator. If the creator also configures a task plug-in, the storage device **420** further comprises a task manager **426** for managing a predetermined task or a user-defined task. The management of tasks can be completed by an event tracker **450** provided by the assistant creation platform **110**.

The interface component **430** of the digital assistant **120** may indicate a response style **432** of the digital assistant **120**, which may be selected by the creator from a response style library **460** provided by the assistant creation platform **110**. In some embodiments, the interface component **430** of the digital assistant **120** may also be configured to comprise a plug-in bar **434** for presenting one or more plug-ins configured by the creator for the digital assistant **120**.

In FIG. **4**, providers of these components of the digital assistant **120** are shown by different blocks and lines, comprising the part provided by the assistant creator (e.g., the user **105**), the part provided by the assistant creation platform **110**, and the part provided by the plug-in development platform (i.e., the plug-in library and its plug-ins). From FIG. **4**, it can be seen that only some configuration information needs to be input by the user to obtain customized development of the digital assistant **120**.

FIG. **5** shows a flowchart of an example process **500** for digital assistant creation in accordance with some embodiments of the present disclosure. The process **500** may be implemented at the assistant creation platform **110**. The process **500** is described below with reference to FIG. **1**.

At block **510**, the assistant creation platform **110**, in response to receiving the creation request, presents a page for creating a digital assistant, the page comprising at least one configuration area for receiving configuration information for the digital assistant. The at least one configuration area comprises: a first configuration area for receiving settings information input in a natural language, the settings information being used to generate a prompt input of a machine learning model, and a response of the digital assistant to a user being determined by the digital assistant based on an output of the model.

At block **520**, the assistant creation platform **110** obtains the configuration information received in the page.

At block **530**, in response to receiving a release request, the assistant creation platform **110** releases the digital assistant based on the configuration information, for use in interaction with the user.

In some embodiments, the at least one configuration area further comprises: a second configuration area for receiving a configuration of at least one processing component, the configuration of the at least one processing component being provided to the model for determining a response to the user.

In some embodiments, releasing the digital assistant comprises: in response to receiving the release request, providing at least one candidate platform, each of the at least one candidate platform supporting execution of the digital assistant; and in response to receiving a determination of a target platform amongst the at least one candidate platform, releasing the digital assistant to the target platform.

In some embodiments, the settings information indicates at least one of: a definition of a response style of the digital assistant, a description of a function supported by the digital assistant, at least one workflow to be performed by the digital assistant, or a definition of at least one response format of the digital assistant.

In some embodiments, the process **500** further comprises: providing a settings information example on the page for guiding a user in providing the settings information of the digital assistant.

In some embodiments, the at least one processing component comprises at least one of: at least one plug-in, each plug-in being configured to perform a corresponding function; at least one workflow to be performed by the digital assistant; at least one dataset which is to be utilized by the digital assistant to determine a response to the user; a definition of at least one type of information to be persistently stored, the at least one type of information being extracted during an interaction between the digital assistant and the user and being stored for a subsequent interaction; or a task plug-in configured to perform a predetermined task or a user-defined task.

In some embodiments, the process **500** further comprises: presenting a respective response configuration control for the at least one plug-in; in response to detecting a trigger operation of a response configuration control of a first plug-in amongst the at least one plug-in, presenting a plurality of candidate response styles associated with the first plug-in; and receiving a selection of a target response pattern amongst the plurality of candidate response styles.

In some embodiments, the at least one configuration area further comprises: a third configuration area for receiving guidance information, the guidance information being presented to a user in response to a detection of the user triggering an interaction with the digital assistant.

In some embodiments, the guidance information comprises at least one of: description information for the digital assistant, or at least one recommendation of a question for the digital assistant.

In some embodiments, the process **500** further comprises: in response to detecting a trigger operation of an information generation control associated with the third configuration area, generating candidate guidance information based at least on the settings information received in the first configuration area; and presenting the candidate guidance information in the third configuration area.

In some embodiments, the at least one configuration area further comprises: a fourth configuration area for receiving a selection of a model, the selected model being invoked to determine a response of the digital assistant to a user.

In some embodiments, the page further comprises: a debugging area for receiving a debugging request for the digital assistant and providing a debugging result for the debugging request, wherein the debugging result is determined based on the received debugging request and current configuration information for the digital assistant in the page.

FIG. **6** shows a schematic block diagram of an example apparatus **600** for digital assistant creation in accordance with some embodiments of the disclosure. The apparatus **600**, for example, may be implemented in or comprised in the assistant creation platform **110**. The various modules/components in the apparatus **600** may be implemented by hardware, software, firmware, or any combination thereof.

As shown in the figure, the apparatus **600** comprises a page presenting module **610** configured to, in response to receiving a creation request, present a page for creating a digital assistant, the page comprising at least one configuration area for receiving configuration information for the digital assistant. The at least one configuration area comprises a first configuration area for receiving settings information input in a natural language, the settings information being used to generate a prompt input of a machine learning model, and a response of the digital assistant to a user being determined by the digital assistant based on an output of the model.

The apparatus **600** further comprises an information obtaining module **620** configured to obtain the configuration information received in the page. The apparatus **600** further comprises an assistant releasing module **630** configured to, in response to receiving a release request, release the digital assistant based on the configuration information, for use in interaction with the user.

In some embodiments, the at least one configuration area further comprises: a second configuration area for receiving a configuration of at least one processing component, the configuration of the at least one processing component being provided to the model for determining a response to the user.

In some embodiments, the assistant releasing module **630** comprises: a candidate platform providing module configured to, in response to receiving the release request, provide at least one candidate platform, each of the at least one candidate platform supporting execution of the digital assistant; and a target platform releasing module configured to, in response to receiving a determination of a target platform amongst the at least one candidate platform, release the digital assistant to the target platform.

In some embodiments, the settings information indicates at least one of: a definition of a response style of the digital assistant, a description of a function supported by the digital assistant, at least one workflow to be performed by the digital assistant, or a definition of at least one response format of the digital assistant.

In some embodiments, apparatus **600** further comprises: an example providing module configured to provide a settings information example on the page for guiding a user in providing the settings information of the digital assistant.

In some embodiments, the at least one processing component comprises at least one of: at least one plug-in, each plug-in being configured to perform a corresponding function; at least one workflow to be performed by the digital assistant; at least one dataset which is to be used by the digital assistant to determine a response to the user; a definition of at least one type of information to be persistently stored, the at least one type of information being extracted during an interaction between the digital assistant and the user and being stored for a subsequent interaction; or a task plug-in configured to perform a predetermined task or a user-defined task.

In some embodiments, the apparatus **600** further comprises: a control presentation module configured to present a respective response configuration control for the at least one plug-in; a response style module configured to, in response to detecting a trigger operation of a response configuration control of a first plug-in amongst the at least one plug-in, present a plurality of candidate response styles associated with the first plug-in; and a style selection module configured to receive a selection of a target response pattern amongst the plurality of candidate response styles.

In some embodiments, the at least one configuration area further comprises: a third configuration area for receiving guidance information, the guidance information being presented to a user in response to a detection of the user triggering an interaction with the digital assistant.

In some embodiments, the guidance information comprises at least one of: description information for the digital assistant, or at least one recommendation of a question for the digital assistant.

In some embodiments, the apparatus **600** further comprises: a candidate guidance generation module configured to, in response to detecting a trigger operation of an information generation control associated with the third configuration area, generate candidate guidance information based at least on the settings information received in the first configuration area; and a candidate guidance presentation module configured to present the candidate guidance information in the third configuration area.

In some embodiments, the at least one configuration area further comprises: a fourth configuration area for receiving a selection of a model, the selected model being invoked to determine a response of the digital assistant to a user.

In some embodiments, the page further comprises: a debugging area for receiving a debugging request for the digital assistant and providing a debugging result for the debugging request, wherein the debugging result is determined based on the received debugging request and current configuration information for the digital assistant in the page.

FIG. **7** shows a block diagram of an example electronic device **700** in which one or more embodiments of the present disclosure may be implemented. It would be appreciated that the electronic device **700** shown in FIG. **7** is only an example and should not constitute any restriction on the function and scope of the embodiments described herein. The electronic device **700** shown in FIG. **7** may comprise or be implemented as the assistant creation platform **110** or the apparatus **600** of FIG. **6**.

As shown in FIG. **7**, the electronic device **700** is in the form of a general electronic device. The components of the electronic device **700** may comprise, but are not limited to, one or more processors or processing units **710**, a memory **720**, a storage device **730**, one or more communication units **740**, one or more input devices **750**, and one or more output devices **760**. The processing units **710** may be actual or virtual processors and can execute various processes according to the programs stored in the memory **720**. In a multi-processor system, multiple processing units execute computer executable instructions in parallel to improve the parallel processing capability of the electronic device **700**.

The electronic device **700** typically comprises a variety of computer storage media. Such media can be any available media that is accessible to the electronic device **700**, comprising but not limited to volatile and non-volatile media, removable and non-removable media. The memory **720** can be volatile memory (such as registers, caches, random access memory (RAM)), nonvolatile memory (such as a read-only memory (ROM), an electrically erasable programmable read-only memory (EEPROM), a flash memory), or some combination thereof. The storage device **730** can be any removable or non-removable medium, and can comprise machine-readable medium, such as a flash drive, a disk, or any other medium which can be used to store information and/or data and can be accessed within the electronic device **700**.

The electronic device **700** may further comprise additional removable/non-removable, volatile/non-volatile storage medium. Although not shown in FIG. **7**, a disk driver for reading from or writing to a removable, non-volatile disk (such as a "floppy disk"), and an optical disk driver for reading from or writing to a removable, non-volatile optical disk can be provided. In these cases, each driver may be connected to the bus (not shown) by one or more data medium interfaces. The memory **720** can comprise a computer program product **725**, which comprises one or more program modules configured to execute various methods or actions of the various embodiments disclosed herein.

The communication unit **740** implements communication with other electronic devices via a communication medium. In addition, functions of components in the electronic device **700** may be implemented by a single computing cluster or multiple computing machines, which can communicate through a communication connection. Therefore, the electronic device **700** may be operated in a networking environment using a logical connection with one or more other servers, a network personal computer (PC), or another network node.

The input device **750** may be one or more input devices, such as a mouse, a keyboard, a trackball, etc. The output device **760** may be one or more output devices, such as a display, a speaker, a printer, etc. The electronic device **700** may also communicate with one or more external devices (not shown) through the communication unit **740** as required. The external device, such as a storage device, a display device, etc., communicate with one or more devices that enable users to interact with the electronic device **700**, or communicate with any device (for example, a network card, a modem, etc.) that makes the electronic device **700** communicate with one or more other computing devices. Such communication may be executed via an input/output (I/O) interface (not shown).

According to example implementation of the present disclosure, there is provided a computer-readable storage medium on which a computer-executable instruction or computer program is stored, wherein the computer-executable instructions are executed by a processor to implement the methods described above.

Various aspects of the present disclosure are described herein with reference to the flow chart and/or the block diagram of the method, the device, the apparatus, and the computer program product implemented in accordance with the present disclosure. It would be appreciated that each block of the flowchart and/or the block diagram and the combination of each block in the flowchart and/or the block diagram may be implemented by computer-readable program instructions.

These computer-readable program instructions may be provided to the processing units of general-purpose computers, special computers, or other programmable data processing devices to produce a machine that generates a device to implement the functions/acts specified in one or more blocks in the flow chart and/or the block diagram when these instructions are executed through the processing units of the computer or other programmable data processing devices. These computer-readable program instructions may also be stored in a computer-readable storage medium. These instructions enable a computer, a programmable data processing device and/or other devices to work in a specific way. Therefore, the computer-readable medium containing the instructions comprises a product, which comprises instructions operable to implement various aspects of the functions/acts specified in one or more blocks in the flowchart and/or the block diagram.

The computer-readable program instructions may be loaded onto a computer, other programmable data processing apparatus, or other devices, so that a series of operational steps can be performed on a computer, other programmable data processing apparatus, or other devices, to generate a computer-implemented process, such that the instructions which execute on a computer, other programmable data processing apparatus, or other devices are operable to implement the functions/acts specified in one or more blocks in the flowchart and/or the block diagram.

The flowchart and the block diagram in the drawings show the possible architecture, functions and operations of the system, the method and the computer program product implemented in accordance with the present disclosure. In this regard, each block in the flowchart or the block diagram may represent a part of a module, a program segment, or instructions, which includes one or more executable instructions for implementing the specified logic function. In some alternative implementations, the functions marked in the block may also occur in a different order from those marked in the drawings. For example, two consecutive blocks may actually be executed in parallel, and sometimes can also be executed in a reverse order, depending on the function involved. It should also be noted that each block in the block diagram and/or the flowchart, and combinations of blocks in the block diagram and/or the flowchart, may be implemented by a dedicated hardware-based system that performs the specified functions or acts, or by the combination of dedicated hardware and computer instructions.

Each implementation of the present disclosure has been described above. The above description provides a number of examples, not exhaustive, and is not limited to the disclosed implementations. Without departing from the scope and spirit of the described implementations, many modifications and changes are obvious to ordinary skill in the art. The selection of terms used in this article aims to best explain the principles, practical application, or improvement of technology in the market of each implementation, or to enable others of ordinary skill in the art to understand the various embodiments disclosed herein.

The invention claimed is:

1. A method for digital assistant creation, comprising:

in response to receiving a digital assistant creation request, presenting a page for creating a digital assistant, wherein the page comprises at least one configuration area for receiving configuration information for the digital assistant, and wherein the at least one configuration area comprises:

a first configuration area for receiving settings information of the digital assistant input in a natural language, wherein the settings information of the digital assistant is used to generate a prompt input of a machine learning model, and wherein a response of the digital assistant provided to a user is determined by the digital assistant based on an output of the machine learning model; and

a second configuration area for receiving a configuration of at least one processing component, wherein the configuration of the at least one processing component is provided to the machine learning model for determining the response of the digital assistant to the user;

obtaining the configuration information for the digital assistant received in the at least one configuration area of the page;

creating the digital assistant based on the configuration information for the digital assistant; and

in response to receiving a digital assistant release request, releasing the digital assistant based on the configuration information for the digital assistant, for use in interaction with the user.

2. The method of claim **1**, wherein releasing the digital assistant comprises:

in response to receiving the digital assistant release request, providing at least one candidate platform, wherein the at least one candidate platform supports execution of the digital assistant; and

in response to receiving a determination of a target platform amongst the at least one candidate platform, releasing the digital assistant to the target platform.

3. The method of claim **1**, wherein the settings information of the digital assistant indicates at least one of:

a definition of a response style of the digital assistant;

a description of a function supported by the digital assistant;

at least one workflow to be performed by the digital assistant; or

a definition of at least one response format of the digital assistant.

4. The method of claim **1**, further comprising:

providing a settings information example on the page for guiding a user in providing the settings information of the digital assistant.

5. The method of claim **1**, wherein the at least one processing component comprises at least one of:

at least one plug-in configured to perform a corresponding function;

at least one workflow to be performed by the digital assistant;

at least one dataset which is to be utilized by the digital assistant to determine the response of the digital assistant to the user;

a definition of at least one type of information to be persistently stored, wherein the at least one type of information is extracted during an interaction between the digital assistant and the user and is stored for a subsequent interaction; or

a task plug-in configured to perform a predetermined task or a user-defined task.

6. The method of claim **5**, further comprising:

presenting a respective response configuration control for the at least one plug-in;

in response to detecting a trigger operation of a response configuration control of a first plug-in amongst the at least one plug-in, presenting a plurality of candidate response styles associated with the first plug-in; and

receiving a selection of a target response pattern amongst the plurality of candidate response styles associated with the first plug-in.

7. The method of claim **1**, wherein the at least one configuration area further comprises:

a third configuration area for receiving guidance information, wherein the guidance information is presented to a user in response to a detection of the user triggering an interaction with the digital assistant.

8. The method of claim **7**, wherein the guidance information comprises at least one of: description information for the digital assistant, or at least one recommendation of a question for the digital assistant.

9. The method of claim **7**, further comprising:

in response to detecting a trigger operation of an information generation control associated with the third configuration area, generating candidate guidance information based at least on the settings information of the digital assistant received in the first configuration area; and

presenting the candidate guidance information in the third configuration area.

10. The method of claim **7**, wherein the at least one configuration area further comprises:

a fourth configuration area for receiving a selection of a model, wherein the selected model is invoked to determine a response of the digital assistant to a user.

11. The method of claim **1**, wherein the page further comprises:

a debugging area for receiving a debugging request for the digital assistant and providing a debugging result for the debugging request,

wherein the debugging result is determined based on the received debugging request and current configuration information for the digital assistant in the page.

12. An electronic device comprising:

at least one processing unit; and

at least one memory being coupled to the at least one processing unit and storing instructions for execution by the at least one processing unit, the instructions, when executed by the at least one processing unit, cause the electronic device to perform operations comprising:

in response to receiving a digital assistant creation request, presenting a page for creating a digital assistant, wherein the page comprises at least one configuration area for receiving configuration information for the digital assistant, and wherein the at least one configuration area comprises:

a first configuration area for receiving settings information of the digital assistant input in a natural language, wherein the settings information of the digital assistant is used to generate a prompt input of a machine learning model, and wherein a response of the digital assistant provided to a user is determined by the digital assistant based on an output of the machine learning model; and

a second configuration area for receiving a configuration of at least one processing component, wherein the configuration of the at least one processing component is provided to the machine learning model for determining the response of the digital assistant to the user;

obtaining the configuration information for the digital assistant received in the at least one configuration area of the page;

creating the digital assistant based on the configuration information for the digital assistant; and

in response to receiving a digital assistant release request, releasing the digital assistant based on the configuration information for the digital assistant, for use in interaction with the user.

13. The electronic device of claim **12**, wherein releasing the digital assistant comprises:

in response to receiving the digital assistant release request, providing at least one candidate platform, wherein the at least one candidate platform supports execution of the digital assistant; and

in response to receiving a determination of a target platform amongst the at least one candidate platform, releasing the digital assistant to the target platform.

14. The electronic device of claim **12**, wherein the settings information of the digital assistant indicates at least one of:

a definition of a response style of the digital assistant;

a description of a function supported by the digital assistant;

at least one workflow to be performed by the digital assistant; or

a definition of at least one response format of the digital assistant.

15. The electronic device of claim 12, wherein the operations further comprise:

providing a settings information example on the page for guiding a user in providing the settings information of the digital assistant.

16. The electronic device of claim 12, wherein the at least one processing component comprises at least one of:

at least one plug-in configured to perform a corresponding function;

at least one workflow to be performed by the digital assistant;

at least one dataset which is to be utilized by the digital assistant to determine the response of the digital assistant to the user;

a definition of at least one type of information to be persistently stored, wherein the at least one type of information is extracted during an interaction between the digital assistant and the user and is stored for a subsequent interaction; or

a task plug-in configured to perform a predetermined task or a user-defined task; and

wherein the operations further comprise:

presenting a respective response configuration control for the at least one plug-in;

in response to detecting a trigger operation of a response configuration control of a first plug-in amongst the at least one plug-in, presenting a plurality of candidate response styles associated with the first plug-in; and

receiving a selection of a target response pattern amongst the plurality of candidate response styles associated with the first plug-in.

17. The electronic device of claim 12, wherein the at least one configuration area further comprises:

a third configuration area for receiving guidance information, wherein the guidance information is presented to a user in response to a detection of the user triggering an interaction with the digital assistant; and

wherein the guidance information comprises at least one of: description information for the digital assistant, or at least one recommendation of a question for the digital assistant.

18. A non-transitory computer-readable storage medium having a computer program stored thereon, the computer program being executable by a processor to perform operations comprising:

in response to receiving a digital assistant creation request, presenting a page for creating a digital assistant, wherein the page comprises at least one configuration area for receiving configuration information for the digital assistant, and wherein the at least one configuration area comprises:

a first configuration area for receiving settings information of the digital assistant input in a natural language, wherein the settings information of the digital assistant is used to generate a prompt input of a machine learning model, and wherein a response of the digital assistant provided to a user is determined by the digital assistant based on an output of the machine learning model; and

a second configuration area for receiving a configuration of at least one processing component, wherein the configuration of the at least one processing component is provided to the machine learning model for determining the response of the digital assistant to the user;

obtaining the configuration information for the digital assistant received in the at least one configuration area of the page;

creating the digital assistant based on the configuration information for the digital assistant; and

in response to receiving a digital assistant release request, releasing the digital assistant based on the configuration information for the digital assistant, for use in interaction with the user.

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